ModelIt!

Teacher's Guide

Designing Organisms That Eat Plastic

Synthetic Biology and Metabolic Engineering for Environmental Remediation

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS1-5, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-5, HS-ETS1-2

Use a model to illustrate how photosynthesis transforms light energy into stored chemical energy; design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Designing Solutions** — Students develop a synthetic biology model integrating enzyme kinetics, metabolic engineering, and environmental conditions to design an optimized biodegradation system.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow / ETS1.B Developing Possible Solutions** — Organisms use energy from chemical reactions to build complex molecules and drive cellular processes; engineering solutions require breaking complex problems into manageable sub-problems.
Crosscutting Concept	**Energy and Matter / Systems and System Models** — Students trace matter and energy flow through the engineered metabolic pathway from plastic polymer to harmless end products, modeling the system across molecular, organism, and environmental scales.

CAST Testing Alignment

MULTIPLE CHOICE:

An engineered plastic-eating bacterium shows excellent degradation in the lab at 37°C but is deployed in ocean water at 15°C. Based on your model, what is the most likely outcome?

CONSTRUCTED RESPONSE:

Using your synthetic biology model, explain why simply engineering a faster enzyme does not solve the ocean plastic problem. Discuss at least three other model components that must be optimized simultaneously and the trade-offs between them.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to synthetic biology and metabolic engineering in the context of plastic bioremediation — one of the most exciting applications of engineered organisms. Biotech skill focus: Synthetic design and systems biology. The discovery of natural plastic-degrading enzymes opened a new frontier in environmental biotechnology, but translating laboratory breakthroughs to real-world solutions requires systems-level synthetic design.

The Plastic Pollution Crisis

Humans have produced approximately 8.3 billion metric tons of plastic since 1950. Of this, only 9% has been recycled, 12% incinerated, and 79% accumulated in landfills or the natural environment. An estimated 8 million tons enter the oceans annually. The Great Pacific Garbage Patch alone contains 80,000 tons across 1.6 million square kilometers. Conventional plastics are synthetic polymers — long chains of repeating chemical units — that no organism evolved to degrade because they didn't exist in nature until the 20th century. PET (polyethylene terephthalate — drink bottles), polyethylene (bags, packaging), polypropylene (food containers), and polystyrene (foam) each have different chemical structures requiring different degradation enzymes. The environmental persistence of these materials ranges from hundreds to thousands of years.

The PETase Discovery

In 2016, researchers at the Kyoto Institute of Technology discovered Ideonella sakaiensis — a bacterium from a Japanese recycling plant that had evolved the ability to break down PET plastic using a novel enzyme called PETase. This was the first confirmed example of a naturally evolved enzyme capable of degrading a synthetic polymer. PETase works by hydrolyzing the ester bonds in PET, producing terephthalic acid and ethylene glycol as breakdown products. A second enzyme, MHETase, completes the degradation of an intermediate compound. However, natural PETase is slow — it takes weeks to significantly degrade a thin plastic film at the enzyme's optimal temperature of 30°C. Protein engineering has created 'supercharged' PETase variants (like FAST-PETase, developed at UT Austin in 2022) that work 5-10x faster and at higher temperatures, but significant challenges remain for real-world deployment.

Metabolic Engineering for Complete Degradation

Breaking PET into terephthalic acid and ethylene glycol is only the first step — the organism must then metabolize these products completely, or they accumulate as pollutants. Complete degradation requires engineering a multi-enzyme metabolic pathway: PETase breaks PET into MHET, MHETase converts MHET to terephthalic acid and ethylene glycol, additional enzymes convert terephthalic acid through protocatechuate to central metabolism (TCA cycle), and ethylene glycol is oxidized to glycolate and enters central metabolism. If any step in this pathway is inefficient, intermediate compounds accumulate — and some of these intermediates (particularly oxidized ethylene glycol derivatives) are more toxic than the original plastic. This is the critical lesson: optimizing one enzyme without optimizing the complete pathway can make the pollution problem worse.

The Containment Challenge

Releasing engineered organisms into the environment is the most controversial aspect of synthetic biology. Containment strategies include: (1) Auxotrophies — engineering the organism to require a synthetic amino acid or nutrient not found in nature, so it dies without laboratory supplementation; (2) Kill switches — genetic circuits that kill the organism in response to environmental signals (temperature change, absence of an inducer chemical); (3) Genetic firewalls — recoding the organism's genetic code so its genes cannot function if transferred to wild microbes; (4) Physical containment — deploying organisms in enclosed bioreactor systems rather than as free-living environmental releases. Each strategy has limitations: kill switches can be mutated, auxotrophies can be bypassed by horizontal gene transfer, and physical containment limits deployment scale. The field of biosafety engineering is dedicated to making containment robust enough for real-world deployment.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten components spanning enzyme biochemistry (Enzyme Activity Level), metabolic engineering (Metabolic Pathway Efficiency, Byproduct Toxicity), organism biology (Organism Growth Rate, pH Tolerance), environmental conditions (Substrate Concentration, Environmental Temperature, Oxygen Requirements), and biosafety (Degradation Rate, Containment Level).

E **Establish Relationships**

Students determine that Substrate Concentration, Environmental Temperature, Oxygen Requirements, Scaffold Structure, and Containment Level are the controllable inputs — while Enzyme Activity Level, Metabolic Pathway Efficiency, Byproduct Toxicity, Organism Growth Rate, pH Tolerance, and Degradation Rate emerge as biological responses to the environmental conditions and genetic engineering decisions.

V **Visualize & Model**

Students build a synthetic biology model connecting enzyme performance to metabolic pathway function to organism-level environmental degradation, visualizing how lab-optimized parameters translate (or fail to translate) to real-world performance.

E **Evaluate Outcomes**

Students run controlled-lab, ocean-deployment, and optimized-field-release scenarios to quantify the performance gap between laboratory and environmental conditions and find the design specifications that balance effectiveness with safety.

R **Revise & Extend**

Students add Biofilm Formation Capacity, Horizontal Gene Transfer Risk, or Multi-Plastic Capability to explore more sophisticated and realistic synthetic biology designs.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every minute, a garbage truck's worth of plastic enters the ocean. We can't clean it up fast enough with machines. But what if we could engineer a living organism that eats it?"*

Watch For

Show the scale of ocean plastic pollution with a visual comparison. Ask: Why can't we just scoop it all out?

Slide 2: Learning Objectives

3 min

Say: *"Today you're synthetic biologists. Your job is to design a living organism from scratch — one that can do something no natural organism has ever done: break down plastic."*

Watch For

Pre-teach synthetic biology and metabolic engineering. Show the Ideonella sakaiensis discovery story as a 1-minute intro.

Slide 3: Big Question

3 min

Say: *"Plastic has only existed for about 70 years. No organism evolved to eat it because it never existed in nature before. But in 2016, scientists found a bacterium in a recycling plant that had EVOLVED to eat PET plastic. What if we could make it better?"*

Watch For

Think-pair-share: Students brainstorm what you'd need to engineer into an organism to make it eat plastic effectively in the ocean. Compile ideas.

Slide 4: LEVER Framework

2 min

Say: *"We're going to design an organism from the enzyme level up — like programming a biological machine with ten interacting parts. This is synthetic biology in action."*

Watch For

Preview LEVER steps. Emphasize the multi-scale design challenge: molecular (enzymes) → cellular (metabolism) → organism (growth) → environmental (deployment).

Slide 5: Activity 1: Components

8 min

Say: *"Four of these ten components are environmental conditions you can choose where to deploy. One is a safety feature you engineer in. Five are biological responses. Sort them."*

Watch For

Guide sorting of external versus internal components. Discuss the containment paradox: you want the organism to spread to plastic but NOT spread everywhere else.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the trap: if you optimize the enzyme to chew up plastic faster but the metabolic pathway can't keep up, you convert solid plastic into dissolved toxins. The cure becomes worse than the disease."*

Watch For

Students trace the complete degradation pathway. Identify where incomplete degradation produces toxic intermediates. Map the containment-effectiveness trade-off.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run your organism in the lab, then drop it in the ocean. Watch what happens to performance. Then try to redesign it for real-world conditions."*

Watch For

Students predict lab-to-field performance drop, then verify with simulations. Challenge: find the design that works in real ocean conditions while maintaining containment.

Slide 8: Discoveries

5 min

Say: *"You just learned the first rule of synthetic biology: what works in the lab almost never works in the real world without major redesign. But the gap is shrinking — and your generation might close it."*

Watch For

Connect model findings to the real FAST-PETase breakthrough. Discuss remaining challenges and timelines for real-world deployment.

Slide 9: STEM Challenge

12 min

Say: *"You have $100 million and the world's best synthetic biology tools. Design the organism — or organism system — that cleans up the Great Pacific Garbage Patch. Go."*

Watch For

Groups design complete biological cleanup systems including organism design, deployment strategy, containment plan, and monitoring system. Present proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Substrate Concentration, Environmental Temperature, Oxygen Requirements, Containment Level
- **INTERNAL:** Enzyme Activity Level, Metabolic Pathway Efficiency, Byproduct Toxicity, Organism Growth Rate, pH Tolerance, Degradation Rate

Relationships

- **Substrate Concentration to Degradation Rate:** POSITIVE (+) — Higher plastic substrate concentration provides more material for the engineered enzyme to process, increasing the overall Degradation Rate up to the enzyme's saturation point.
- **Enzyme Activity Level to Degradation Rate:** POSITIVE (+) — More efficient enzymes process more plastic molecules per second, directly increasing the rate at which plastic substrate is converted to breakdown products.
- **Environmental Temperature to Enzyme Activity Level:** POSITIVE (+) up to optimum, then NEGATIVE (-) — Enzyme activity increases with temperature up to the optimum (30-65°C for most PETase variants) then drops sharply as the enzyme denatures — critical for ocean deployment where temperatures are far below optimum.
- **Metabolic Pathway Efficiency to Byproduct Toxicity:** NEGATIVE (-) — More efficient metabolic pathways convert plastic breakdown products completely to harmless end products (CO2, biomass), reducing the accumulation of toxic intermediate compounds.
- **Organism Growth Rate to Degradation Rate:** POSITIVE (+) — Faster organism proliferation increases the total enzyme concentration in the environment, amplifying the system-wide Degradation Rate as the microbial population expands.
- **Containment Level to Organism Growth Rate:** NEGATIVE (-) — Stronger containment features (kill switches, auxotrophies) impose metabolic costs on the organism and limit its ability to proliferate freely, reducing the population density achievable in the environment.

Key Discoveries

- The gap between laboratory enzyme performance and real-world degradation is enormous — enzymes optimized for lab conditions lose 80-95% of their activity in environmental conditions (cold ocean water, variable pH, limited oxygen)
- Incomplete plastic breakdown can generate toxic intermediates — optimizing enzyme activity without optimizing the complete metabolic pathway can make the pollution problem WORSE by converting plastic into soluble toxins
- The containment-effectiveness paradox is a fundamental challenge — organisms need to spread to where plastic accumulates, but uncontrolled spread of engineered organisms poses ecological risks we cannot fully predict
- Temperature is the critical limiting factor — most plastic-degrading enzymes have evolved or been engineered for warm conditions (30-65°C), but ocean temperatures average 15°C and much plastic accumulates in cold deep-sea environments

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We can just release bacteria into the ocean and the plastic problem is solved"	Releasing engineered organisms into the ocean faces enormous challenges the model reveals: cold temperatures slow enzyme activity by 80-95%, nutrient-poor ocean water limits organism growth, UV radiation damages DNA, and the organisms must somehow find and attach to widely dispersed plastic particles. Additionally, the sheer scale of ocean plastic (80,000+ tons in the Pacific alone) would require organism populations of astronomical size. Bioremediation is a tool that complements other approaches — it's not a magic bullet.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Engineered organisms are inherently dangerous and should never be released"	Risk assessment of engineered organisms must be evidence-based, not fear-based. Billions of engineered bacteria are already used safely in industrial fermentation (producing insulin, enzymes, vitamins). The key is containment engineering — multiple redundant safety systems that prevent the organism from surviving outside its intended deployment context. The model's Containment Level component shows that well-designed safety features can dramatically reduce environmental risk while maintaining useful function.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plastic is just a chemistry problem — biology is too slow"	While chemical recycling can process plastic faster than biological degradation, it requires high temperatures, harsh chemicals, and centralized infrastructure. Biology offers something chemistry cannot: self-replication and self-deployment. An engineered organism can reproduce, spread to plastic-contaminated sites, and adapt to local conditions — potentially providing continuous, self-sustaining cleanup in remote locations where chemical processing infrastructure doesn't exist.